When we say 'battery has no charge', it means, that the battery has lost ability to provide energy or provide potential difference to the electrons in the circuit. When we say 'mobile is charging', it implies that the battery is receiving energy from AC power supply and not electrons.

2.1.3 Microscopic model of current

Consider a conductor with area of cross section A and let an electric field $\vec{E}$ be applied to it from right to left. Suppose there are n electrons per unit volume in the conductor and assume that all the electrons move with the same drift velocity $\vec{v}_d$ as shown in Figure 2.5.

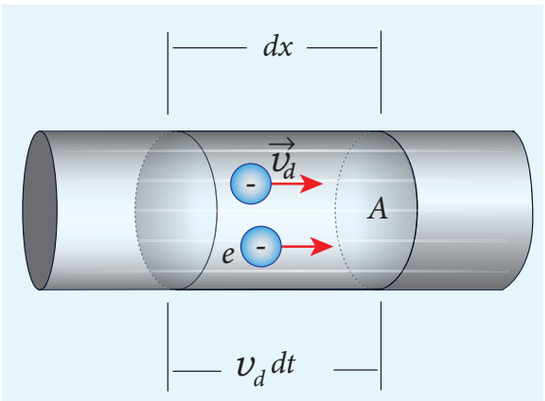


Figure 2.5 Microscopic model of current

The drift velocity of the electrons = v_d

If the electrons move through a distance dx within a small interval of dt, then

$$v_d = \frac{dx}{dt}\,; \quad dx = v_d\,dt \tag{2.7}$$

Since A is the area of cross section of the conductor, the electrons available in the volume of length dx is

= volume × number of electrons per unit volume

$$= A\,dx \times n \tag{2.8}$$

Substituting for dx from equation (2.7) in (2.8)

$$= (A\ v_d\,dt\ )\ n$$

Total charge in the volume element dQ = (charge) × (number of electrons in the volume element)

$$dQ = (e)(A v_d dt)n$$

Hence the current $I = \frac{dQ}{dt}$

$$I = ne\,A v_d \tag{2.9}$$

Current density (J)

The current density (J) is defined as the current per unit area of cross section of the conductor.

$$J = \frac{I}{A}$$

The S.I unit of current density is $\frac{\text{A}}{\text{m}^2}$ (or) A m^{-2}

$J = \frac{neAv_d}{A}$ (from equation 2.9)

$$J = nev_d \tag{2.10}$$

The above expression is valid only when the direction of the current is perpendicular to the area A. In general, the current density is a vector quantity and it is given by

$$\vec{J} = ne\vec{v}_d$$

Substituting $\vec{v}_d$ from equation (2.4)

$$\vec{J} = -\frac{n \cdot e^2 \tau}{m}\vec{E} \tag{2.11}$$

$$\vec{J} = -\sigma\vec{E}$$

But conventionally, we take the direction of (conventional) current density as the direction of electric field. So the above equation becomes

$$\vec{J} = \sigma\vec{E} \tag{2.12}$$

where $\sigma = \frac{ne^2\tau}{m}$ is called conductivity. The equation (2.12) is called microscopic form of ohm's law.

12th Physics_Vol-1_Unit 2_English Medium.indd 85 12-12-2022 17:21:30